JORDAN DOWDY

Ph.D. Candidate (ABD), Electrical and Computer Engineering

Telos Lab, Louisville Automation & Robotics Research Institute (LARRI) | University of Louisville

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jdowdy05.github.io | Google Scholar | GitHub | LinkedIn

RESEARCH INTERESTS

Reinforcement learning and representation learning for adaptive robot control, emphasizing sim-to-real locomotion, privileged information transfer, and deployable learning systems for legged and humanoid robots.

EDUCATION

Ph.D. Candidate, Electrical and Computer Engineering — August 2024–Present

University of Louisville | Louisville, Kentucky

Advisor: Jean Chagas Vaz, Ph.D. | *ABD (All But Dissertation).*

M.S., Electrical Engineering — January 2022–August 2024

University of Louisville | Louisville, Kentucky

Advisor: Dan O. Popa, Ph.D. | **Thesis:** *Adaptive Robot Collaboration Using Robotic Skin and Motion Similarity*

B.S., Physics; Minor in Mathematics — May 2021

Bellarmine University | Louisville, Kentucky

RESEARCH EXPERIENCE

Graduate Research Assistant — August 2024–Present

Telos Lab, LARRI, University of Louisville | Advisor: Jean Chagas Vaz, Ph.D.

- Introduced a novel neural-network architecture (STRIDE), a privileged knowledge-distillation framework using Barlow-style latent alignment and corrective transformer residuals for sim-to-real blind quadruped locomotion.
- Developed a token-compressed temporal transformer that fuses sparse, low-cost time-of-flight terrain data with proprioception for sim-to-real ROBOTIS OP3 locomotion over uneven terrain.
- Developed a VR imitation-learning pipeline for cube stacking in Isaac Sim that retargets demonstrations into reusable joint-space datasets with post-collection sensor re-rendering and Isaac Mimic expansion.
- Zero-shot transferred several Isaac Sim/Isaac Lab locomotion policies to ROBOTIS OP3 and Unitree Go1 hardware.
- Co-developed a full-body VR tele-loco-manipulation stack for miniature humanoid robots that combines reinforcement-learned locomotion with upper-body teleoperation.

Graduate Research Assistant (M.S.) — January 2022–August 2024

LARRI, University of Louisville | Advisor: Dan O. Popa, Ph.D.

- Created Cartesian Segment Online Dynamic Time-Warping (SODTW), a novel segment-based online DTW variant for real-time motion-similarity estimation and speed-adaptive robot imitation.
- Independently developed the OctoCan tactile-guided physical human-robot interaction system with ARMA/RLS calibration and neuroadaptive control; co-authored its second-stage validation paper.

Research Intern — May 2020–August 2020

Norton Cancer Institute

- Contributed to an apparatus designed to improve uniform radiation-dose delivery for pediatric cancer patients.

Undergraduate Research Assistant — Fall 2018–May 2022

Physics Department, Bellarmine University | Advisor: Akhtar Mahmood, Ph.D.

- Robotics Lab (2018–2020): Built Arduino robots integrating servo motors, sensors, and circuitry.
- Astrophysics/OBARO (2020–2022): Built a Python/GMM pipeline to classify objects in Rubin/LSST PhoSim FITS imagery.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, LabVIEW, Bash

Robotics and Simulation: ROS 1/ROS 2, NVIDIA Isaac Sim, Isaac Lab, MuJoCo

Robot Learning and Scientific Computing: PyTorch, TensorFlow, NumPy, SciPy, pandas

Deployment and Systems: ONNX, NVIDIA Jetson, Docker, Linux, GStreamer, Open MPI

Engineering Tools: Altium Designer, Fusion 360, Git, LaTeX

PUBLICATIONS

- **STRIDE: Self-Supervised Transformer Residual Corrective Connections with Privileged Knowledge Distillation in Locomotion**
  J. Dowdy and J. C. Vaz
  Manuscript in preparation for submission to IEEE Robotics and Automation Letters (RA-L), 2026.
- **Real-Time Neuroadaptive Control with Tactile Calibration for Physical Human-Robot Interaction**
  A. Prakash, M. A. Hanafy, **J. Dowdy**, and D. O. Popa
  Electronics, accepted for publication, 2026. Preprint.
- **Loco-Loco-RL: Low-Cost Terrain Mapping for Humanoid Locomotion with Reinforcement Learning**
  J. Dowdy, G. Reizian, and J. C. Vaz
  IEEE National Aerospace and Electronics Conference (NAECON), 2026.
- **Learning to Stack: Cube-Stacking Imitation Learning from Virtual Reality Demonstrations**
  G. Reizian†, **J. Dowdy†**, and J. C. Vaz
  IEEE National Aerospace and Electronics Conference (NAECON), 2026. †Equal contribution.
- **Towards Torque-Driven Reinforcement Learning for Quadruped Locomotion**
  J. Dowdy and J. C. Vaz
  IEEE/SICE International Symposium on System Integration (SII), 2026, pp. 1259–1264. Oral presentation.
- **Towards Miniature Humanoid Tele-Loco-Manipulation Using Virtual Reality and Reinforcement Learning**
  N. Kosanovic, **J. Dowdy**, and J. C. Vaz
  IEEE-RAS International Conference on Humanoid Robots (Humanoids), 2025, pp. 1233–1240.
- **Isaac Sim-to-Real: Reinforcement Learning Based Locomotion for Quadrupeds**
  J. Dowdy and J. C. Vaz
  IEEE International Conference on Automation Science and Engineering (CASE), 2025, pp. 2194–2199. Oral presentation.
- **Adaptive Motion Imitation for Robot Assisted Physiotherapy Using Dynamic Time Warping and Recurrent Neural Network**
  A. Ashary, M. M. Rayguru, **J. Dowdy**, N. Taghavi, and D. O. Popa
  International Conference on PErvasive Technologies Related to Assistive Environments (PETRA), 2024, pp. 55–62.
- **Adaptive Robot Collaboration Using Robotic Skin and Motion Similarity**
  J. Dowdy
  M.S. thesis, Department of Electrical and Computer Engineering, University of Louisville, 2024.

ACADEMIC SERVICE

Teaching and Instructional Support

- **Lab Assistant, Computational Physics** | Spring 2022. Department of Physics, Bellarmine University, Louisville, Kentucky. Supported laboratory instruction in Linux, MPI, troubleshooting, and parallel-code execution; guided students in building a 12-node Linux Beowulf cluster with Open MPI.

Conference Reviewing

- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS); IEEE-RAS International Conference on Humanoid Robots (Humanoids); IEEE International Conference on Automation Science and Engineering (CASE); IEEE International Conference on Robotics and Automation (ICRA); and IEEE/SICE International Symposium on System Integration (SII).

Journal Reviewing

- IEEE Robotics and Automation Letters (RA-L).

Research Mentoring

- Mentored five undergraduate researchers—two during the M.S. and three during the Ph.D.—on robotics research.

HONORS AND PROFESSIONAL MEMBERSHIPS

- **Graduate Student Researcher Fellowship** | University of Louisville, 2024–Present.
- **Student Member** | IEEE and IEEE Robotics and Automation Society (RAS).